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```
% Yeshwantrao Chavan College of Engineering
% Department of Electronics and Telecommunication Engineering
% Session EVEN 2023-24
% Name of Student:Rutuja Saharkar
clc;
clear all;
close all;
```

Experiment 6:Spatial Domain Filtering (High Pass Filtering)

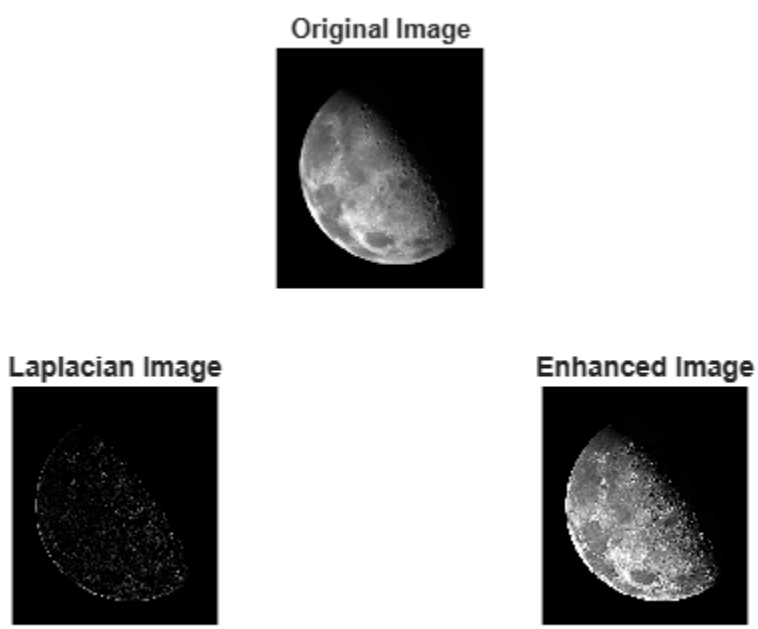
Function for filtering

```
%Filter Function
%function[g]=myfilter(f,mask)
%[m,n]=size of mask
%a=(m-1)/2;
%b=(n-1)/2;
%f1=padarray(f,[a b],"replicate","both")
%[r c]=size of f1
%g=zeros(r,c);
%for x=a+1:r-a
%for y=b+1:c-b
%for s=-a:a
%for t=-b:b
%g(x,y)=g(x,y)+f(x+s,y+t)*mask(s+a+1,t+b+1)
%end
%end
%end
%end
```

```
%g=g(a+1:r-a,b+1:c-b);
%end
```

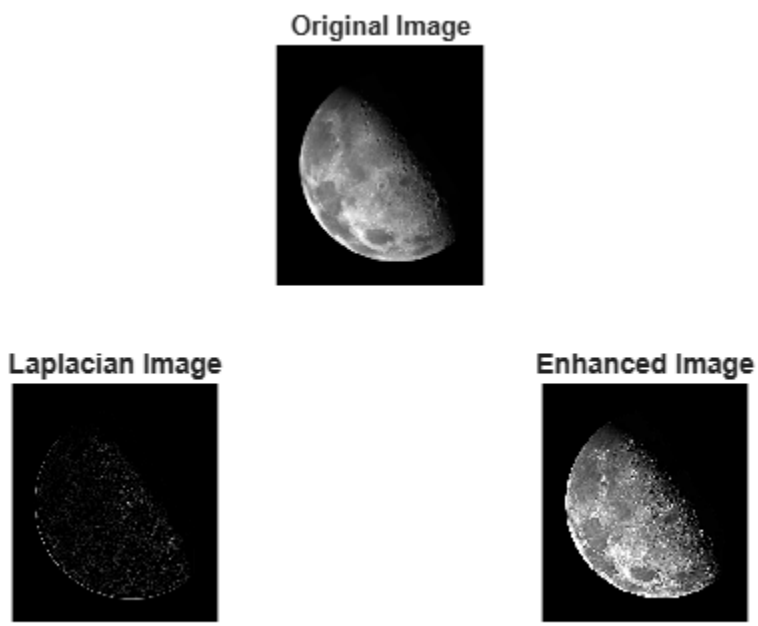
Excercise 1

```
%Perform image enhancement using Laplacian
% Read the image
f = imread('Fig16.tif');
% Convert to double precision using im2double
f = im2double(f);
% Declare isotropic laplacian mask of size 3X3 with centered postive
laplacian_mask = [-1 -1 -1;-1 8 -1;-1 -1 -1];
% Call filtering function to obtain laplacian image
laplacian_image = myfilter(f, laplacian_mask);
% Add laplacian image with orginal to get enhaced image
enhanced_image = f + laplacian_image;
% Display original image, Laplacian image and enhanced image in 2x2 window
subplot(2,2,[1,2]),
imshow(f),
title('Original Image');
subplot(2,2,3),
imshow(laplacian_image),
title('Laplacian Image');
subplot(2,2,4),
imshow(enhanced_image),
title('Enhanced Image');
```



Excercise 2

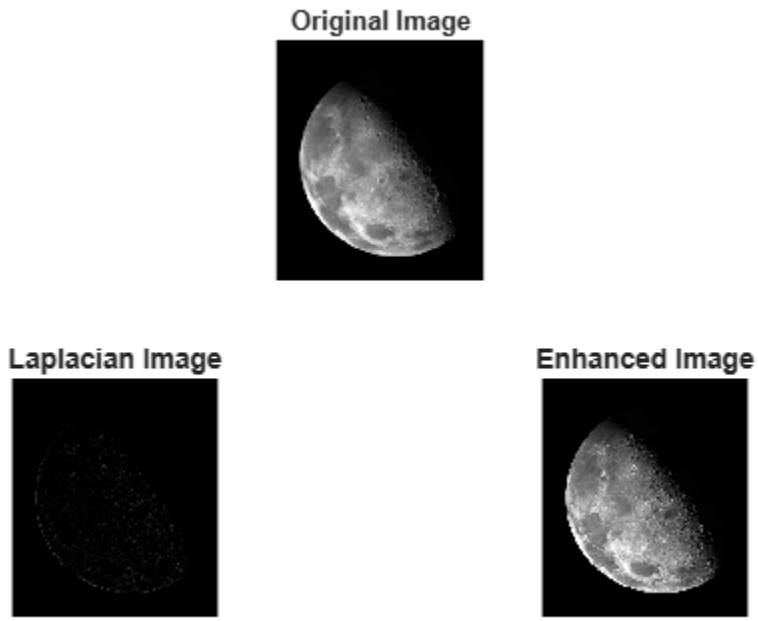
```
%Perform image enhancement using Laplacian
% Read the image
f = imread('Fig16.tif');
% Convert to double precision using im2double
f = im2double(f);
% Declare isotropic laplacian mask of size 3X3 with centered negative
laplacian_mask = [1 1 1;1 -8 1;1 1 1];
% Call filtering function to obtain laplacian image
laplacian_image = myfilter(f, laplacian_mask);
% subtract laplacian image with orginal to get enhaced image
enhanced_image = f - laplacian_image;
% Display original image, Laplacian image and enhanced image in 2x2 window
subplot(2,2,[1,2]),
imshow(f),
title('Original Image');
subplot(2,2,3),
imshow(laplacian_image),
title('Laplacian Image');
subplot(2,2,4),
imshow(enhanced_image),
title('Enhanced Image');
```



Excercise 3

```
%Perform image enhancement using Laplacian
% Read the image
f = imread('Fig16.tif');
```

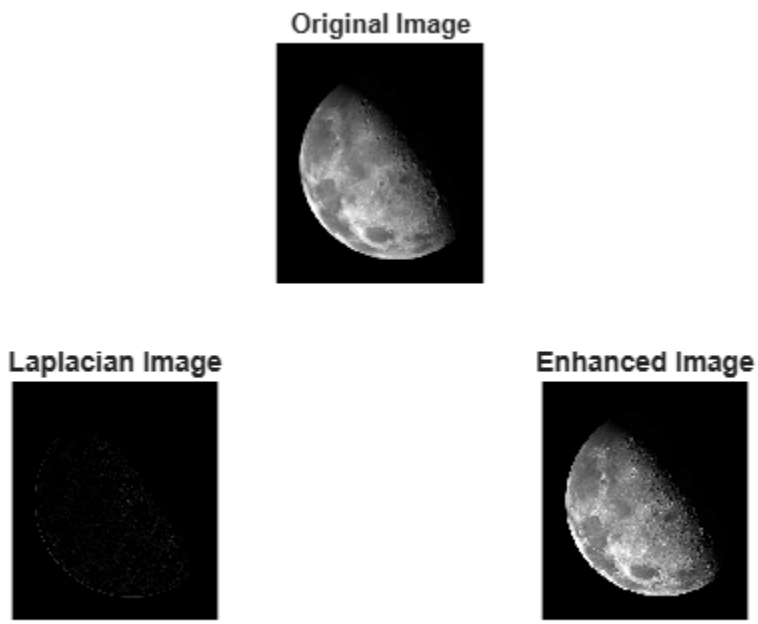
```
% Convert to double precision using im2double
f = im2double(f);
% Declare laplacian mask isotropic in +-90 of size 3X3 with centered postive
laplacian_mask = [0 -1 0;-1 4 -1;0 -1 0];
% Call filtering function to obtain laplacian image
laplacian_image = myfilter(f, laplacian_mask);
% Add laplacian image with orginal to get enhaced image
enhanced_image = f + laplacian_image;
% Display original image, Laplacian image and enhanced image in 2x2 window
subplot(2,2,[1,2]),
imshow(f),
title('Original Image');
subplot(2,2,3),
imshow(laplacian_image),
title('Laplacian Image');
subplot(2,2,4),
imshow(enhanced_image),
title('Enhanced Image');
```



Excercise 4

```
%Perform image enhancement using Laplacian
% Read the image
f = imread('Fig16.tif');
% Convert to double precision using im2double
f = im2double(f);
% Declare laplacian mask isotropic in +-90 of size 3X3 with centered negative
laplacian_mask = [0 1 0;1 -4 1;0 1 0];
% Call filtering function to obtain laplacian image
laplacian_image = myfilter(f, laplacian_mask);
```

```
% subtract laplacian image with orginal to get enhaced image
enhanced_image = f - laplacian_image;
% Display original image, Laplacian image and enhanced image in 2x2 window
subplot(2,2,[1,2]),
imshow(f),
title('Original Image');
subplot(2,2,3),
imshow(laplacian_image),
title('Laplacian Image');
subplot(2,2,4),
imshow(enhanced_image),
title('Enhanced Image');
```



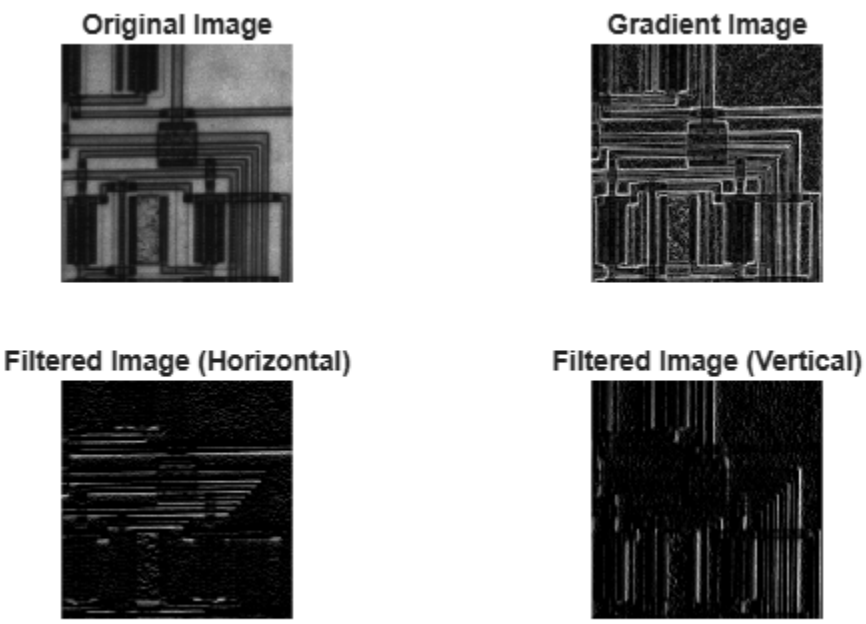
Conclusion:

The Laplacian of an image highlights regions of rapid intensity change and is an example of a second order or a second derivative method of enhancement. It is particularly good at finding the fine details of an image. Any feature with a sharp discontinuity will be enhanced by a Laplacian operator.

Excercise 5

```
%Perform image enhancement using Gradient
% Read the image
f = imread("circuit.tif");
% Convert to double precision using im2double
f = im2double(f);
% Declare Prewit Mask in Horizontal and Vertical Direction
prewitt_mask_x = [-1 -1 -1; 0 0 0; 1 1 1];
prewitt_mask_y = [-1 0 1; -1 0 1; -1 0 1];
```

```
% Call filter function to obtain filtered images
filtered_image_x = myfilter(f,prewitt_mask_x);
filtered_image_y = myfilter(f,prewitt_mask_y);
% Obtain Gradient image using G=|Gx|+|Gy|
gradient_image = abs(filtered_image_x) + abs(filtered_image_y);
% Display the orginal image and Gradient images in 2X2 Window
subplot(2, 2, 1),
imshow(f),
title('Original Image');
subplot(2, 2, 3),
imshow(filtered_image_x),
title('Filtered Image (Horizontal)');
subplot(2, 2, 4),
imshow(filtered_image_y),
title('Filtered Image (Vertical)');
subplot(2, 2, 2),
imshow(gradient_image),
title('Gradient Image');
```



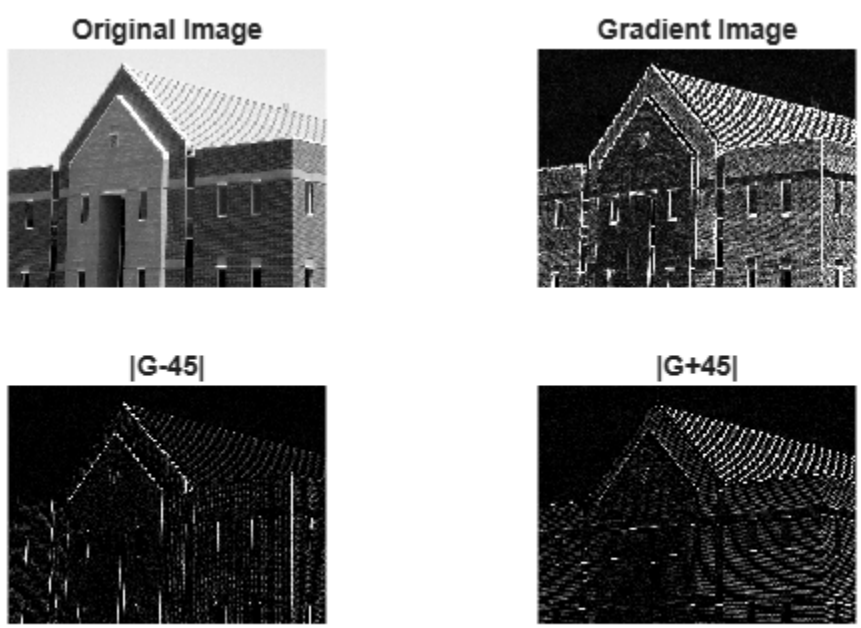
Conclusion

Prewitt operator is used for edge detection in an image.Wherever there is a sudden change in pixel intensities, an edge is detected by the mask. Since the edge is defined as the change in pixel intensities, it can be calculated by using differentiation. Prewitt mask is a first-order derivative mask.Prewitt operator provides us two masks one for detecting edges in the horizontal direction and another for detecting edges in a vertical direction.

Excercise 6

```
%Perform image enhancement using Gradient
% Read the image
```

```
f = imread("Fig17.tif");
% Convert to double precision using im2double
f = im2double(f);
% Declare Sobel Mask in +-45 Direction
sobel_mask_45=[-1 0 1;-2 0 2;-1 0 1];
sobel_mask_minus45=[-1 -2 -1;0 0 0;1 2 1];
% Call filter function to obtain filtered images
filtered_image_45 = myfilter(f,sobel_mask_45);
filtered_image_minus45 = myfilter(f,sobel_mask_minus45);
% Obtain Gradient image using G=|G45|+|G-45|
gradient_image = abs(filtered_image_45) + abs(filtered_image_minus45);
% Display the orginal image and Gradient images in 2X2 Window
subplot(2, 2, 1),
imshow(f),
title('Original Image');
subplot(2, 2, 3),
imshow(filtered_image_45),
title('|G-45|');
subplot(2, 2, 4),
imshow(filtered_image_minus45),
title('|G+45|');
subplot(2, 2, 2),
imshow(gradient_image),
title('Gradient Image');
```



Conclusion

Like the Prewitt operator Sobel operator is also used to detect two kinds of edges in an image i.e Vertical direction and Horizontal direction. The difference between Sobel and Prewitt Operator is that in Sobel operator the

coefficients of masks are adjustable according to our requirement provided they follow all properties of derivative masks.

Excercise 7

```
%Perform image enhancement using unsharpmasking and highboost filtering
% Read the image
I = imread("Fig18.tif");
% Convert to double precision using im2double
I = im2double(I);
% Generate mask for unsharp masking
avg_mask=ones(3,3)/9;
f=[0 0 0;0 1 0;0 0 0];
g_mask=imsubtract(f,avg_mask);
unsharp_mask=imadd(f,g_mask);
% Generate mask for highboost filtering with k>1 (Select any value)
hbstgmsk=imadd(f,10*g_mask);
% Call filter function to obtain unsharp masking image
unsharp_image=myfilter(I,unsharp_mask);
% Call filter function to obtain high boost filtered image
hbst_image=myfilter(I,hbstgmsk);
% Display the orginal image, unsharp masked and highboost filtered image
subplot(3, 1, 1),
imshow(I),
title('Original Image');
subplot(3, 1, 2),
imshow(unsharp_image),
title('unsharp masking image');
subplot(3, 1, 3),
imshow(hbst_image),
title('high boost filtered image');
```

Conclusion

Unsharp masking is defined as obtaining a highpass filtered image by subtracting from the given image a lowpass filtered version of itself while high-boost filtering generalizes it by multiplying the input image by a constant, say K≥1.

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